SEGMENTATION

What is Image Segmentation?

- **Image segmentation** is the process of identifying and outlining specific structures within a medical image.
- In a **CT (Computed Tomography) scan**, different tissues appear with different shades of grey based on their density.
- Segmentation separates structures of interest, such as bones, from the surrounding tissues.
- This process transforms medical images into data that computers can analyse and use for 3D modelling.

What can we learn from medical images?

- CT scans provide detailed information about the shape and structure of bones inside the body.
- By identifying anatomical structures, researchers can measure shape, size, and spatial relationships.
- Medical image analysis helps scientists study anatomy, movement, injury, and disease.

Why is segmentation useful?

- Segmentation allows researchers to create accurate **3D models** of bones and other anatomical structures.
- These models can be used in biomechanics research, surgical planning, and medical device design.
- Image segmentation is an important step in many modern medical imaging and artificial intelligence applications.
- By turning images into digital models, we can better understand the structure and function of the human body.